THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 5: **Measurement of Length and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If the end of a ruler is broken, we should: 1

(a) start from the broken end and read directly
(b) measure from the middle of the ruler
(c) use the ruler upside down
(d) start measuring from a full mark like 1 cm and subtract it

2. The motion of a spinning top about its own axis is: 1

(a) random **(b)** periodic only **(c)** rotational **(d)** linear

3. Which of the following is a non-standard unit of length? 1

(a) handspan **(b)** kilometre **(c)** centimetre **(d)** metre

4. The correct way to measure the length of a pencil is to place the zero mark of the ruler: 1

(a) anywhere **(b)** at one end of the pencil
(c) at the far end **(d)** in the middle

5. The motion of Earth around the Sun is: 1

(a) random **(b)** circular (orbital)
(c) linear **(d)** rotational only

6. The ancient unit 'cubit' was the length from the elbow to the: 1

(a) wrist **(b)** tip of the middle finger
(c) shoulder **(d)** knee

7. Measurements should always be written with: 1

(a) a drawing **(b)** no number **(c)** units **(d)** a name

8. Which of the following is a random motion? 1

(a) a butterfly flying **(b)** a pendulum
(c) a train on a track **(d)** a clock's hand

9. Why are standard units needed? 1

(a) to measure only small objects
(b) to get the same measurement whoever measures
(c) to make every object longer
(d) to avoid using a scale

10. The motion of a car moving on a straight road is: 1

(a) random **(b)** periodic only

(c) linear (rectilinear) **(d)** circular

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Name any two types of motion with an example each. 2

12. Convert 5 cm to millimetres. 2

13. Convert 3500 m into kilometres. 2

14. Define periodic motion with an example. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A rope measures 6.4 m. Express its length in centimetres and millimetres. 3

16. Explain relative motion with the example of a passenger in a train. 3

17. Describe the measurement of a curved line with a thread. 3

OR

Explain how to measure the thickness of a sheet of paper using a book. 3

18. Why do trees seem to move when we travel in a car? 3

OR

What is the SI unit of length? Write the relationship with cm, mm and km. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Discuss the need for a standard unit of length and describe the SI unit. 5

OR

Explain the periodic motion of a pendulum, measure the time of 20 oscillations and find the time of one if 20 oscillations take 40 s. 5

20. Explain parallax error and describe the correct method for measuring the length of a pencil. 5

OR

Classify the following: spinning top, moving car, swing, flying bird, hand of a clock. 5

END OF QUESTION PAPER Total: 40 Marks

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